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Relationships between plasma depletion and subsolar reconnection

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Abstract. Observations in the subsolar magnetosheath show that the plasma depletion layer (PDL) is less pronounced for southward than for northward IMF. Since subsolar plasma depletion indicates pile-up of magnetic flux, the degree of plasma depletion should depend on the relative rates of flux transport via subsolar reconnection and flux advection by the solar wind flow. To identify the factors affecting plasma depletion, we consider the ratio $D = E_r/E_{sw}$ where E_r is the reconnection electric field and E_{sw} is the imposed solar wind electric field. For a quasi-perpendicular subsolar bow shock D can be expressed in terms of magnetosheath parameters. Since PDL formation is suppressed when the subsolar bow shock is quasi-parallel, we restrict attention to quasi-perpendicular conditions. We show that D increases with increasing reconnection efficiency, magnetic shear at the magnetopause, and the magnetosheath magnetic field but decreases with increasing total perpendicular pressure (particle plus magnetic field) in the magnetosheath. By combining observations of the subsolar quasi-perpendicular magnetosheath from AMPTE/IRM and AMPTE/CCE we verify that the degree of plasma depletion is inversely correlated with D. Furthermore, we show that the greater prevalence of plasma depletion in the CCE data implies that the reconnection efficiency is lower for the CCE events and specifically that the reconnection efficiency depends roughly inversely on magnetosheath β, for $\beta > 1$. Finally, the results show that all of the changes brought about by plasma depletion act to increase E_r so that some subsolar reconnection is likely to occur even for low magnetic shear. Thus the percentage of the magnetosheath magnetic field that participates in reconnection near the subsolar region does not act as a rectifier but remains positive for all shear angles, decreasing monotonically as the magnetic shear at the subsolar magnetopause changes from high to low shear. This implies that the equatorward polar cap convection cells observed during northward IMF and conventionally thought to be driven by a viscous interaction may be due at least in part if not wholly to subsolar low shear reconnection.

1. Introduction

Momentum and energy transfer from the solar wind to the magnetosphere is believed to occur near the subsolar magnetopause via reconnection when the solar wind magnetic field is southward. The magnetosheath plasma imposed on the magnetosphere is therefore of great importance for this transport process. Near the subsolar region the magnetosheath flow can give rise to a region of decreased density and increased magnetic field strength called the plasma depletion layer (PDL) when the subsolar bow shock is quasi-perpendicular. When a PDL forms, it is the plasma and field in this layer, rather than the magnetosheath proper, that is imposed on the subsolar magnetopause.

There is some variation in the reported prevalence of the PDL. *Anderson and Fuselier* [1993], hereinafter P1, reported a PDL for all orientations of the magnetosheath magnetic field, although the density decrease and field increase were smallest when the field was southward. By contrast, *Phan et al.* [1994], hereinafter P2, found that a PDL was clearly evident only when the magnetosheath field was aligned within 60° of the magnetospheric field. Both P1 and P2 interpreted the dependence of PDL prevalence on IMF orientation as evidence for reconnection flows at the subsolar magnetopause, but the difference in PDL occurrence for high shear between P1 and P2 has not been explained. The issue of PDL formation for southward IMF is important since its formation indicates that mag-

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netic flux is piling up against the magnetopause despite the reconnection flows and hence that the reconnection electric field has saturated.

The data sets used in P1 and P2 were different. The data of P1 are from the Active Magnetosphere Particle Tracers Explorers (AMPTE) Charge Composition Explorer (CCE), while the data of P2 are from the AMPTE Ion Release Module (IRM) [cf. *Acuña et al.*, 1985]. The geocentric distances of apogee for these spacecraft were significantly different: 8.8 R_E for CCE and 18.8 R_E for IRM. The nominal subsolar magnetopause standoff distance is 11 R_E [*Roelof and Sibeck*, 1993], so all of the CCE magnetosheath encounters occurred for high solar wind Alfvén Mach number, whereas IRM crossed the magnetosheath under generally more typical solar wind conditions.

The aim of this paper is to understand why CCE and IRM observed a difference in occurrence of plasma depletion. In section 2 we present a model of the relationship between plasma depletion and reconnection to identify the variables that control PDL formation. In section 3 we describe how we combined the two data sets and restricted the data to quasi-perpendicular upstream conditions. Section 4 presents a joint analysis of the databases of P1 and P2 to test the predictions of our model and to deduce the dependence of the reconnection electric field on magnetic shear and magnetosheath plasma β. The implications of the results are discussed in section 5.

The results indicate that the difference in PDL formation between the CCE and IRM data sets is due largely to a reduced reconnection efficiency for the CCE cases attributed to a threefold increase in magnetosheath β for the CCE events relative to the IRM cases. We also point out that the effects of plasma depletion enhance the probability and rate of subsolar reconnection, so that when the magnetosheath conditions are unfavorable for subsolar reconnection, the formation of the PDL acts to restore some subsolar reconnection. Principally, this is due to the fact that the magnetosheath field strength in the PDL increases as the IMF turns northward, implying that subsolar component reconnection remains allowed, albeit at a reduced rate, for all IMF orientations except exactly northward. We therefore propose that subsolar reconnection and plasma depletion should be regarded as dynamically coupled. When conditions in the magnetosheath are unfavorable for subsolar reconnection, plasma depletion occurs and acts to adjust conditions adjacent to the magnetopause so that some subsolar reconnection is likely to occur for almost all upstream conditions and IMF orientations. From this perspective, four-cell convection for IMF $B_z > 0$ can be understood as resulting from the combined effects of cusp reconnection and simultaneous weak subsolar reconnection without recourse to a viscous interaction.

2. Simple Model of Plasma Depletion and Reconnection

In this section we consider the related effects of solar wind and reconnection flows to determine the effect that various parameters should have on PDL formation. Magnetic flux is convected earthward by the solar wind at the rate given by the electric field

$$E_{sw} = B_{sw} V_{sw} \tag{1}$$

where B_{sw} is the magnitude of the solar wind magnetic field transverse to the Earth-Sun line, and V_{sw} is the solar wind velocity. Because the PDL forms preferentially for a quasi-perpendicular subsolar bow shock (e.g., P1), we do not consider the parallel shock case and assume that the solar wind magnetic field is draped in an orderly way. Note that we are not just interested in the southward component of the solar wind magnetic field. Magnetic flux is convected into the reconnection line at a rate given by the reconnection electric field

$$E_r = B_r V_r \tag{2}$$

where B_r is the magnitude of the magnetic field that participates in reconnection and V_r is the inflow velocity from the magnetosheath side into the reconnection line. If we consider the ratio of the E_r and E_{sw}

$$D = \frac{E_r}{E_{sw}} \tag{3}$$

we expect that D is always less than 1. If $D = 0$, then no reconnection occurs and all of the solar wind magnetic field is diverted around the magnetopause via MHD flow. Under these circumstances a plasma depletion layer will form [*Midgley and Davis*, 1963; *Lees*, 1964; *Zwan and Wolf*, 1976]. As D increases, reconnection is able to take up some of the solar wind magnetic flux and the PDL will be correspondingly diminished.

We would like to determine how D depends on the solar wind velocity, density, and magnetic field strength and orientation. We note that V_r depends on the Alfvén speed in the magnetosheath and on the reconnection efficiency, k,

$$V_r = k V_{A,r} \tag{4}$$

where $V_{A,r} = B_r/\sqrt{(\mu_0 \rho_s)}$ and ρ_s is the magnetosheath mass density [e.g., *Sonnerup*, 1974]. That is, $V_{A,r}$ is the magnetosheath Alfvén speed using the component of the sheath field that participates in merging. Typically, k is assumed to be around 0.1, and although it is not known how k depends on the properties of the magnetosheath plasma it is expected to go to zero for high β [*Sonnerup*, 1974].

Denoting the magnetic field in the magnetosheath and in the magnetosphere as $\mathbf{B}_s$ and $\mathbf{B}_m$, respectively, B_r is the projection of $\mathbf{B}_s$ along the $\mathbf{B}_s - \mathbf{B}_m$ direction:

$$B_r = \mathbf{B}_s \cdot \frac{\mathbf{B}_s - \mathbf{B}_m}{|\mathbf{B}_s - \mathbf{B}_m|} \tag{5}$$

When $B_r < 0$, reconnection is geometrically impossible. This is the condition originally discussed by *Sonnerup* [1974]. Because of the PDL, the magnetosheath field strength adjacent to the magnetopause, B_s^*, is not independent of the shear angle, $\Delta\theta$, between $\mathbf{B}_s$ and $\mathbf{B}_m$. In both P1 and P2 it was found that B_s^* approaches B_m as $\Delta\theta$ goes to zero. Therefore we do not assume that B_r necessarily becomes negative, although it must go to zero for $\Delta\theta = 0$. We write $B_r = S(\Delta\theta) B_s$ where $S(0) = 0$, $S(\pi) = 1$ and $S(\Delta\theta)$ increases monotonically with $\Delta\theta$. Then (2) and (4) give

$$E_r = \frac{k(\beta_s) S(\Delta\theta)^2 B_s^2}{\sqrt{\mu_0 \rho_s}} \tag{6}$$

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To express D in terms of solar wind parameters, we must make some assumption about the shock. If we restrict attention to strong, quasi-perpendicular shocks we can write $B_s = \eta B_{sw}$ and $\rho_s = \eta \rho_s$ where η is close to 4 [e.g., *Kennel et al.*, 1985]. The downstream β is determined principally by the solar wind Alfvén Mach number, M_{Asw}. As M_{Asw} increases, η does not change much and approaches 4 asymptotically, but β_s increases like M_{Asw}^2 because the shock heating increases in proportion to the upstream kinetic energy. We therefore write $k(M_{Asw})$ and have

$$D = \frac{k(M_{Asw}) S(\Delta\theta)^2 B_{sw} \eta^{1.5}}{\sqrt{\mu_0 P_{ram}}} \tag{7}$$

where P_{ram} is the solar wind ram pressure, $P_{ram} = \rho_{sw} V_{sw}^2$.

This result suggests that the PDL is most favored under low magnetic shear, high solar wind ram pressure conditions consistent with those of P1 and P2. There are two additional important dependencies predicted by (7). First, large B_{sw} should inhibit PDL formation because although higher B_{sw} implies a higher solar wind magnetic flux transport, it also implies both more flux transport via reconnection and a higher reconnection rate because of the higher Alfvén speed in the sheath. Second, the PDL may be favored when M_{Asw} is large because k decreases with increasing β_s which increases with increasing M_{Asw}.

To compare this model with magnetosheath observations, we express D in terms of quantities measured in the sheath. For the cases discussed below, the ram pressure in the sheath is negligible relative to the perpendicular (magnetic plus thermal) pressure, $P_{\perp Tot,s}$, so we have

$$D = \frac{k(\beta_s) S(\Delta\theta)^2 B_s \eta^{0.5}}{\sqrt{\mu_0 P_{\perp Tot,s}}} \tag{8}$$

The parameters in (8) correspond to values in the magnetosheath proper, upstream of the PDL if present. This is the primary expression used below for comparison with the observations.

3. Combining the IRM and CCE Databases

To combine the IRM and CCE databases we first ensured that the events selected from each data set corresponded as nearly as possible to the same region of space sampled under comparable shock geometries. Each event is defined as two adjacent time intervals, one from the magnetosheath proper, denoted 1, and the other from the magnetosheath near the magnetopause, denoted 2. The magnetosheath field in 2 was used to evaluate the magnetic shear, $\Delta\theta$, between the magnetosheath near the magnetopause and the magnetosphere. We adopt the magnetosheath proper/PDL intervals of P1 (CCE) and refer to them here as regions 1 and 2, respectively. For the P2 events (IRM), we associate the magnetosheath proper with the time interval 10 to 20 min before (after) the magnetopause on inbound (outbound) crossings. The corresponding time interval for the near magnetopause region was chosen to be 0 to 5 min. We follow P2 and categorize events according to the magnetic shear. Cases with $\Delta\theta < 60°$ are called low shear, and those with $\Delta\theta > 60°$ we call high shear.

CCE: Equivalence of PDL and Near Magnetopause Region

For CCE the low radial spacecraft velocities required that the region identification in P1 be based on local plasma signatures. In P1 the PDL was shown to be correlated with left-handed proton cyclotron waves occurring above the local He^{2+} gyrofrequency, whereas the magnetosheath proper was indicated by mirror mode like fluctuations. The correspondence of this evolution in spectral signatures from the magnetosheath proper to the plasma depletion layer has been confirmed both observationally and theoretically [*Anderson et al.*, 1994; *Denton et al.*, 1994]. The departure of the ion temperature anisotropy below the mirror instability threshold in the PDL necessarily implies that such a mode changeover occurs and is associated with the magnetosheath proper/PDL transition [P2; *Phan et al.*, 1996b].

It is important to consider whether adopting the events of P1 constitutes preselection of the CCE data set against events without plasma depletion layers. First, we note that P1 showed that for quasi-perpendicular, $Q_\perp$, upstream bow shock conditions, 77% of all magnetopause crossings were associated with transition into a PDL. Second, we consider those CCE magnetopause crossings for which the magnetic field and plasma parameters displayed well-behaved, monotonic profiles in the magnetosheath. For these crossings the upstream conditions did not change suddenly, thus giving the best picture of the magnetosheath structure near the magnetopause for the CCE events. These crossings have been presented previously (Plates 1 and 2 of P1) and displayed the above-mentioned spectral characteristics of proton cyclotron waves regardless of the magnetic shear. Thus the presence of the PDL for southward magnetosheath fields is a persistent feature in the CCE observations. Adopting the events of P1 therefore does not exclude a major population of high shear cases without a PDL because such conditions are in the minority in the CCE data set.

Upstream Shock Geometry

We restricted the IRM events of P2 to ensure that the upstream shock geometries were comparable. The upstream shock condition is important because as found in P1, the PDL forms preferentially for $Q_\perp$ upstream conditions. The events in P1 identified as PDL all displayed an absence of energetic, >10 keV, He^{2+} indicating connection to a $Q_\perp$ bow shock. Corresponding He^{2+} data are not available for the IRM database of P2, so instead we used the R parameter of *Lin et al.* [1991] to discriminate between $Q_\perp$ and quasi-parallel, $Q_\parallel$, upstream shock conditions. This parameter is

$$R \equiv \frac{1}{N} \left| \sum_{\Delta t} \frac{\vec{\mathbf{B}}_i}{B_i} \right| \tag{9}$$

where the $\mathbf{B}_i$ are sequential vector samples of the magnetic field, B_i is the field magnitude, Δt is the summation time interval, and N is the number of samples in the sum. If $R \sim 1$, the field orientation is well ordered and nearly constant but $R < 1$ indicates a variable field direction. As *Luhmann et al.* [1986] and *Lin et al.* [1991] discussed, well-ordered field directions are characteristic of $Q_\perp$ up-

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stream conditions, whereas $Q_{\|}$ conditions are associated with variable downstream field orientations.

To ensure that the intervals from P1 and P2 used here correspond to similar upstream shock orientations, we evaluated R for the CCE events and used only those IRM events that gave R values in the same range as the CCE events. To evaluate R for the CCE events, we used a single time interval for each event which spanned both regions 1 and 2 and for the IRM events we used the full 20-min interval sunward of the magnetopause crossing. We evaluated R during each event at intervals separated by $\Delta t/4$ and calculated the average R value for the event, denoted R_{avg}. Values for Δt of both 2 min and 5 min were used and gave essentially the same results. For the CCE events, no R value was less than 0.7 and R_{avg} was larger than 0.94 in every case. We therefore selected only those IRM events which satisfied $R > 0.7$, and $R_{avg} > 0.9$. Of the original 38 events in P2, 25 events met these criteria and were used in this analysis.

The two data sets therefore both correspond to quasi-perpendicular bow shock conditions. Although the subsolar magnetosheath is not highly sensitive to bow shock geometry for an IMF within 45° of the Y–Z GSE plane [*Kennel et al.*, 1985; *Wu*, 1992] there may be some variation in PDL severity within the data sets due to the range of subsolar shock orientation within the quasi-perpendicular regime. However, temporal variations in solar wind conditions will probably be a more important complicating factor. In any case, we expect that shock geometry variations and temporal change effects will tend to average out in each data set. We are therefore interested only in average trends.

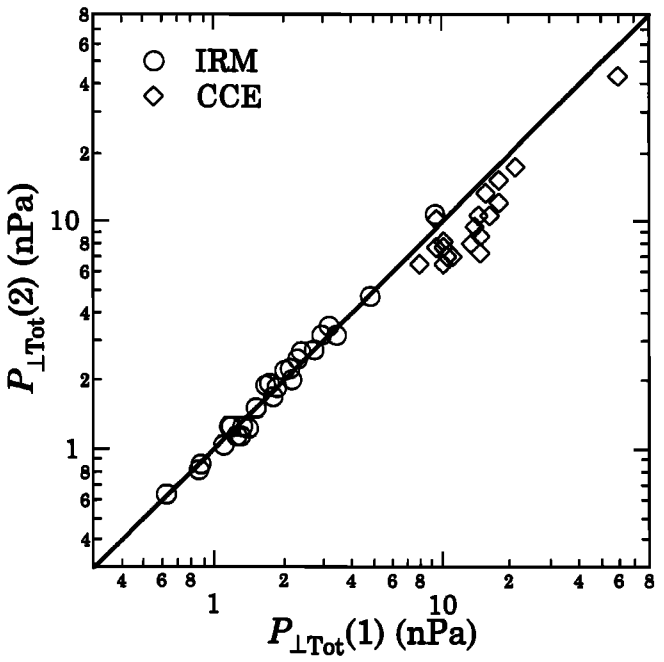

Figure 1. Total perpendicular static pressure, $P_{\perp Tot} = P_{\perp part} + B^2/2\mu_0$, observed near the magnetopause region 2 plotted versus $P_{\perp Tot}$ observed in the magnetosheath proper. That $P_{\perp Tot}(1) = P_{\perp Tot}(2)$ for the IRM events supports the interpretation that IRM traversed a stationary structure. For CCE, $P_{\perp Tot}(2)/P_{\perp Tot}(1) = 0.72$ on average, indicating that the magnetopause was about 0.5 R_E more earthward when CCE sampled region 1 than when CCE sampled region 2.

4. Observational Results and Model Comparison

Total Pressure Consistency Checks

Comparing the total pressure in regions 1 and 2 is useful both to compare conditions for the CCE and IRM events and to examine whether there are any systematic variations in upstream conditions between the time intervals used for the two regions. Such a systematic difference is expected for the CCE events because the CCE magnetosheath observations occurred near CCE apogee. Thus whether the magnetosheath proper or PDL was sampled by CCE depended on the imposed dynamic pressure, and we expect that the magnetosphere must be somewhat more compressed for CCE to sample the magnetosheath proper than for it to sample the PDL.

To estimate the dynamic pressure, we use the total static pressure normal to the magnetopause, $P_{\perp Tot} = P_{\perp part} + B^2/2\mu_0$, that is, the sum of magnetic and perpendicular particle pressures. The IRM ram plasma velocities given in P2 indicate that the ram pressure for the IRM events is ~0.02 nPa (using a typical magnetosheath density of 20 cm^{-3}), which is negligible compared to $P_{\perp Tot}$. We therefore ignore the magnetosheath ram pressure and assume that the ram pressure is negligible relative to $P_{\perp Tot}$ for the CCE events as well. The field line curvature near the magnetopause is also a negligible effect, so we consider only the scalar magnetic pressure for approximating force balance. In the case of IRM, all ions are interpreted as H^+, whereas for CCE the H^+ and He^{2+} contributions are evaluated separately and added to give $P_{\perp part}$. For CCE the He^{2+} contribution to the pressure was ~10%, which is small relative to the difference in pressure between the CCE and IRM data sets.

In Figure 1 we plot $P_{\perp Tot}(2)$ versus $P_{\perp Tot}(1)$. As expected, the pressures during the IRM and CCE events were very different. For IRM the average $P_{\perp Tot}(1)$ was 1.7 (2.8) nPa for low (high) shear whereas for CCE the average $P_{\perp Tot}(1)$ was 9.6 (12.4) nPa for low (high) shear. The comparison of $P_{\perp Tot}$ in regions 1 and 2 is also significant. For the IRM events, $P_{\perp Tot}(2) \sim P_{\perp Tot}(1)$, as expected, if IRM traversed a stationary structure. For the CCE events $P_{\perp Tot}(2) < P_{\perp Tot}(1)$, as expected, if 2 is closer to the magnetopause than 1, since lower $P_{\perp Tot}$ brings CCE closer to the magnetopause when CCE is in the magnetosheath. The average ratio, $P_{\perp Tot}(2)/P_{\perp Tot}(1)$, for CCE is 0.72 which provides a measure of the scale length for the magnetosheath proper to PDL transition. The CCE radial distance at apogee is 8.8 R_E, so if we take this for the radial distance for 2, $r(2)$, then $r(1)$ for the same dynamic pressure will be given by $r(2)/r(1) \sim 0.72^{1/6} = 0.95$ so that $r(1) - r(2) \sim 0.5$ R_E. Thus the magnetosheath proper/PDL transition occurs over about 0.5 R_E.

Average Sheath and Inferred Solar Wind Conditions

To quantify the differences between the CCE and IRM events, we compare the average magnetosheath conditions for the data sets. Averages, standard deviations, and minimum and maximum values for $\beta(1)$, $B(1)$, $n(1)$, $P_{\perp Tot}(1)$, and the Alfvén velocity, $V_A(1)$, are given in Table 1. In

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Table 1. Average IRM and CCE Parameters in the Magnetosheath Proper Denoted in the Text as Region 1

	IRM		CCE	
	Average ± Standard Deviation	(min, max)	Average ± Standard Deviation	(min, max)
$\beta(1)$	2.5 ± 1.4	(0.51, 5.57)	8.0 ± 4.1	(2.71, 14.3)
$B(1)$, nT	39.7 ± 19.5	(17.2, 119.0)	67.4 ± 16.5	(40.1, 120.9)
$n(1)$, cm^{-3}	21.2 ± 18.0	(5.5, 95.2)	166 ± 108	(43.9, 537)
$P_{\perp\mathrm{Tot}}(1)$, nPa	2.15 ± 1.69	(0.61, 9.26)	15.8 ± 11.2	(7.9, 59.5)
V_A, km/s	210 ± 85	(90, 395)	130 ± 50	(70, 250)

addition to the large difference in $P_{\perp\mathrm{Tot}}(1)$, $\beta(1)$ is roughly three times higher for the CCE events than for the IRM events. The difference in $\beta(1)$ is due principally to the eightfold higher density for the CCE events, which more than compensates for the higher magnetic field strength of the CCE events. Owing to the huge densities for the CCE events, $V_A(1)$ is lower for the CCE than for the IRM events.

It is also useful to estimate the solar wind conditions during the events. We can infer the upstream solar wind conditions to ~20% on the basis of observations in the magnetosheath proper. We estimate the field strength, B_{sw}, solar wind flow speed, V_{sw}, and proton density, n_{sw}, from parameters in region 1 as follows. Across the shock, B_{sw} and n_{sw} increase by a factor, η, which depends on the upstream Mach number and on the angle between $\mathbf{B}_{sw}$ and the shock normal, θ_{Bn}. We then have, $B_{sw} = B(1)/\eta$ and $n_{sw} = n(1)/\eta$. The solar wind ram pressure is approximately $n_{sw}m_pV_{sw}^2$, where m_p is the proton mass, and we have ignored He^{2+}. The ram pressure is also given by $P_{\perp\mathrm{Tot}}(1)$, so we obtain

$$V_{sw} = \sqrt{\frac{\eta P_{\perp\mathrm{Tot}}(1)}{m_p n(1)}} \tag{10}$$

From B_{sw}, V_{sw}, and n_{sw} we then calculate E_{sw} and the Alfvén Mach number, M_{Asw}. Note that E_{sw} and M_{Asw} vary only as $\eta^{-1/2}$ so that these estimates are not very sensitive to η. Moreover, for quasi-perpendicular shocks the jump conditions are not highly sensitive to θ_{Bn} and for strong shocks $\eta \sim 4$ [*Kennel et al.*, 1985]. Table 2 presents the statistics of the inferred solar wind parameters assuming $\eta = 4$.

The inferred solar wind parameters for IRM are very close to average solar wind conditions. The differences in n_{sw}, B_{sw}, and V_{Asw} carry over directly from the magnetosheath values. The inferred solar wind velocities, V_{sw}, are comparable however. As a result, M_{Asw} for the CCE events is higher than for the IRM events. For both data sets, M_{Asw} is always larger than 3.6 and the average M_{Asw} is greater than 5, indicating that these events occurred for strong shocks, consistent with the assumption regarding the jump conditions, that is, $\eta = 4$.

Measure of Plasma Depletion

To measure the degree of plasma depletion for a given event, we use the ratio of plasma β values in region 1 and 2, $\beta(2)/\beta(1)$. As shown in P1 and P2, the transition from the magnetosheath proper to the PDL corresponds to the simultaneous variation of a number of parameters: the density decreases, the average magnetic field strength increases, and the ion temperature decreases. These changes all contribute to a decrease in the plasma β, so we adopt this parameter as a measure of the degree of plasma depletion. To remove variations in $\beta(1)$ from event to event, we restrict attention to the ratio $\beta(2)/\beta(1)$. When $\beta(2)/\beta(1) \sim 1$, little plasma depletion in region 2 is indicated, whereas $\beta(2)/\beta(1) << 1$ indicates significant depletion in region 2.

Since the CCE events do not correspond to traversal of a stationary structure, $\beta(1)$ and $\beta(2)$ for the CCE events are not directly comparable in the sense that they necessarily correspond to somewhat different upstream conditions. Specifically, we know that $P_{\perp\mathrm{Tot}}(2) < P_{\perp\mathrm{Tot}}(1)$. Hence for the CCE events we estimate $\beta(2)$ corresponding to conditions near the magnetopause when CCE was in region 1, denoted as $\beta(2)'$. Because the measured $\beta(2)$ is probably lower than $\beta(2)'$, the measured $\beta(2)$ gives a lower limit value for $\beta(2)'$. An upper limit for $\beta(2)'$ is obtained by assuming

Table 2. Inferred Solar Wind Parameters for IRM and CCE Events Evaluated Assuming a Factor of 4 Increase in Density and Magnetic Field Magnitude Across the Shock

	IRM		CCE	
	Average ± Standard Deviation	(min, max)	Average ± Standard Deviation	(min, max)
B_{sw}, nT	9.9 ± 4.9	(4.3, 30)	16.9 ± 4.1	(10, 30)
n_{sw}, cm^{-3}	5.3 ± 4.3	(1.4, 24)	42 ± 27	(11, 135)
V_{sw}, km/s	520 ± 140	(290,780)	490 ± 95	(380, 720)
V_{Asw}, km/s	100 ± 40	(45, 200)	65 ± 25	(35, 125)
M_{Asw}	5.3 ± 1.0	(3.6, 7.1)	8.3 ± 1.9	(5.6, 11.5)

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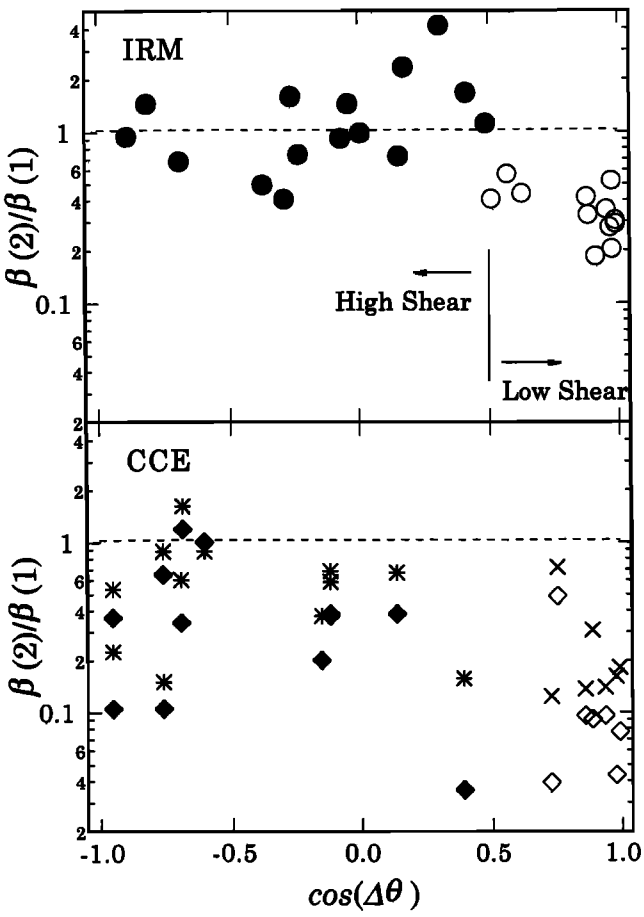

Figure 2. Plot of $\beta(2)/\beta(1)$ versus the cosine of the magnetic shear angle for IRM and CCE. For IRM $\beta(2)/\beta(1)$ are indicated with circles, open for low shear, filled for high shear. For CCE, diamonds indicate $\beta(2)/\beta(1)$, open for low shear, filled for high shear and $\beta(2)^*/\beta(1)$ are shown with crosses for low shear and asterisks for high shear. See text for definitions of $\beta(1)$, $\beta(2)$, and $\beta(2)^*$. Plasma depletion is indicated by $\beta(2)/\beta(1) < 1$. For both IRM and CCE, plasma depletion becomes more prominent as the magnetic shear decreases.

that the change in $P_{\perp Tot}$, $\Delta P_{\perp Tot} = P_{\perp Tot}(1) - P_{\perp Tot}(2)$, required to move CCE from region 2 to region 1, all goes into particle pressure in region 2. That is, $\beta(2)' \leq (P_{part}(2) + \Delta P_{\perp Tot})/P_B(2) = \beta(2)^*$.

In the plots of $\beta(2)/\beta(1)$ given below we indicate IRM data with circles, open for low shear, filled for high shear. For CCE we indicate $\beta(2)/\beta(1)$ by diamonds, open for low shear, filled for high shear; and $\beta(2)^*/\beta(1)$ by crosses for low shear and asterisks for high shear. Note that high and low shear correspond to $\cos(\Delta\theta) < 0.5$ and $\cos(\Delta\theta) > 0.5$, respectively. When referring to $\beta(2)/\beta(1)$ for CCE, we implicitly also refer to $\beta(2)^*/\beta(1)$ unless otherwise stated.

Plasma Depletion, Shear, Field Strength, and Pressure

Figure 2 shows $\beta(2)/\beta(1)$ plotted versus $\cos(\Delta\theta)$ separately for CCE and IRM. Table 3 gives the average values of $\beta(2)/\beta(1)$ for high and low shear as well as the ratio of the high and low shear averages. Both IRM and CCE observe increased plasma depletion with decreasing shear.

Table 3. Average $\beta(2)/\beta(1)$ for IRM and CCE Events Together With the Ratio for High to Low Magnetic Shear

	Average $\beta(2)/\beta(1)$		
	Low Shear	High Shear	High/Low
CCE	0.13	0.42	3.3
IRM	0.35	1.34	3.8

This confirms the expectation from the model that shear should have a prominent influence on plasma depletion. However, for a given shear, the plasma depletion is greater for CCE than for IRM. At high shear, most of the CCE events display $\beta(2)/\beta(1) < 0.4$ whereas all of the high shear IRM events have $\beta(2)/\beta(1) > 0.4$. For low shear, plasma depletion for CCE is greater than for IRM and almost all CCE events have $\beta(2)/\beta(1) < 0.1$, whereas the IRM low shear events have $\beta(2)/\beta(1) > 0.15$. These trends are reflected in the average $\beta(2)/\beta(1)$ values. The variations of $\beta(2)/\beta(1)$ with shear for the CCE and IRM data sets are essentially the same, possibly indicating that the shear dependence of plasma depletion is due to the same mechanism but that the different conditions caused uniformly greater plasma depletion for the CCE cases.

The model predicts that plasma depletion occurs even for high shear if either the pressure is high or the reconnection efficiency is low. Even under these conditions, however, increasing $B(1)$ should reduce D, reduce plasma depletion, and hence give increased $\beta(2)/\beta(1)$. To test for this effect we plot $\beta(2)/\beta(1)$ versus $B(1)$ in Figure 3 for the high shear CCE events. We see that $\beta(2)/\beta(1)$ increases as $B(1)$ increases, indicating reduced plasma depletion as predicted by the model.

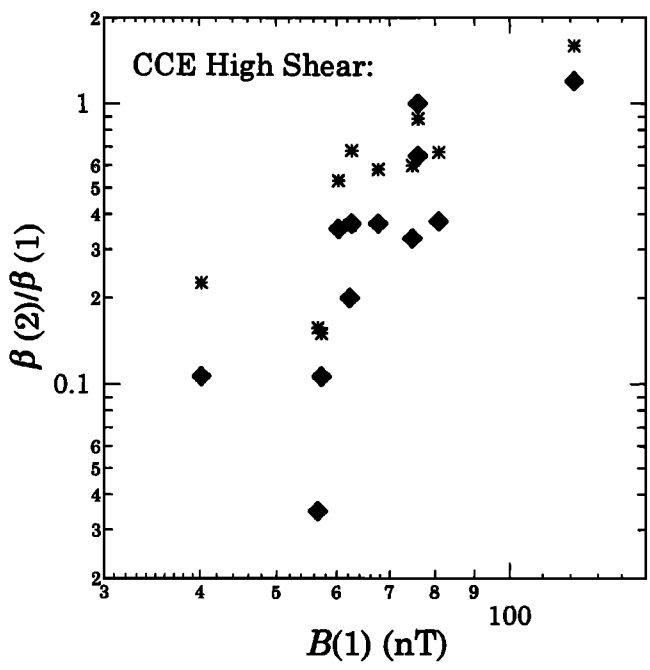

Figure 3. Plot of $\beta(2)/\beta(1)$ versus $B(1)$ for the CCE high shear, $\cos(\Delta\theta) < 0.5$, events showing that plasma depletion is diminished and $\beta(2)/\beta(1)$ increases as the magnetic field strength in the magnetosheath proper, $B(1)$, increases.

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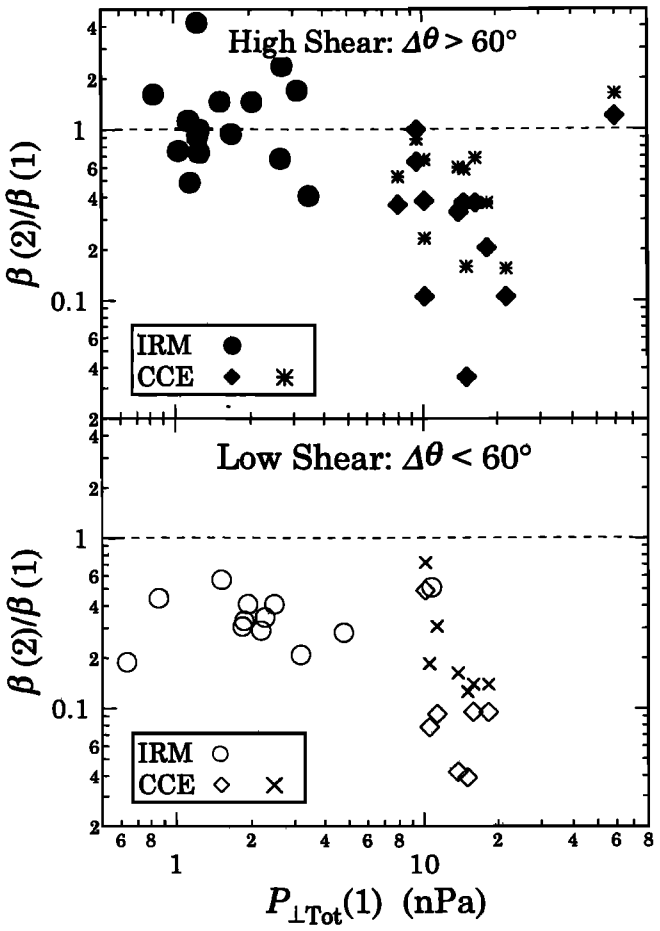

Figure 4. Plot of $\beta(2)/\beta(1)$ versus $P_{\perp Tot}(2)$ for high ($\cos(\Delta\theta) < 0.5$) and low ($\cos(\Delta\theta) > 0.5$) magnetic shear conditions. For both high and low magnetic shear, plasma depletion becomes stronger as the dynamic pressure increases.

Figure 4 shows $\beta(2)/\beta(1)$ plotted versus $P_{\perp Tot}(1)$ separately for high and low magnetic shear. For both high and low shear it is clear that plasma depletion becomes more severe as the applied pressure increases. For low shear the plasma depletion, which is clearly apparent for pressures of ~2 nPa, becomes even stronger as the pressure increases so that $\beta(2)/\beta(1)$ decreases by about a factor of five as $P_{\perp Tot}(1)$ increases from 2 nPa to 10 nPa. A similar behavior occurs for high shear. Below about 3 to 4 nPa, no plasma depletion is evident, but $\beta(2)/\beta(1)$ decreases by about a factor of 2 to 3 for pressures of 10 nPa. This shows that for a given magnetic shear the plasma depletion becomes stronger as the pressure increases, also in agreement with the model. Note, however, that Figure 4 reflects the combined effects of $P_{\perp Tot}$ and β, through k, and because $P_{\perp Tot}$ is positively correlated with β, the effect of pressure on plasma depletion is exaggerated in this plot.

Reconnection Magnetic Field

In order to test whether the difference in $P_{\perp Tot}$ between the CCE and IRM events accounts for the difference in plasma depletion, we need to characterize the shear function, $S(\Delta\theta)$, that appears in (8). This can be done using (5) given $\Delta\theta$ and the field strengths in the magnetosheath just outside the magnetopause, B_s^*, and in the magnetosphere, B_m. For IRM we determined B_m by averaging the field strength over the +1 to +2 min time interval corresponding to periods inside the magnetopause and just inside the LLBL. We cannot measure B_m for the CCE events because not every event is associated with a magnetopause crossing. Moreover, for the CCE events, magnetopause crossings do not correspond to the same upstream conditions as when CCE observed the magnetosheath proper. Therefore we must use only the IRM data to estimate $S(\Delta\theta)$.

The fact that the magnetic field enhancement in the PDL becomes stronger as the magnetosheath field turns northward implies that component merging may be allowed for nearly all orientations of the magnetosheath field. Figure 5 illustrates this effect. As sketched in Figure 5a, component merging can occur only when there exists a coordinate system in which the field in the magnetosheath, $\mathbf{B}_s$, and in the magnetosphere, $\mathbf{B}_m$, have antiparallel components. This requirement gives the condition $B_s^*/B_m > \cos(\Delta\theta)$ for which reconnection may occur [*Sonnerup*, 1974]. If B_s^* is independent of $\Delta\theta$, then as the

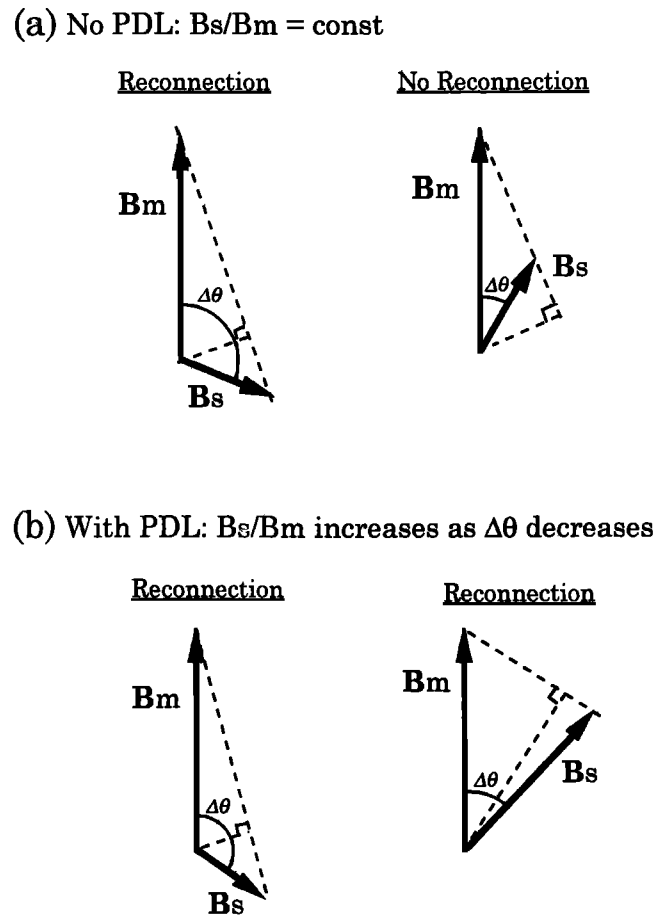

Figure 5. Sketch of relative geometry of magnetospheric magnetic field, vertical arrows, and the magnetosheath magnetic field illustrating the *Sonnerup* [1974] conditions for reconnection; (a) the turn off of reconnection with shear for magnetosheath field strength independent of $\Delta\theta$; and (b) allowed reconnection for all shear angles if the effects of plasma depletion are included.

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shear decreases a point is reached where $B_s^*/B_m = \cos(\Delta\theta)$ and for smaller $\Delta\theta$ there exists no coordinate system in which the fields have antiparallel components and component merging cannot occur (Figure 5a). Because of the PDL, however, B_s^* is not independent of $\Delta\theta$ but increases as the sheath field turns northward. Figure 5b indicates how this affects the merging condition. As $\Delta\theta$ decreases, B_s^* increases, and even as $\Delta\theta$ approaches zero, $B_s^*/B_m > \cos(\Delta\theta)$ may still be satisfied if $B_s^*/B_m \sim 1$. In fact, B_s^*/B_m does approach 1 when $\Delta\theta$ is small [P1; P2; *Phan et al.*, 1996b], implying that the component merging condition can be satisfied in the subsolar region for nearly all shear angles. The quantity of magnetic flux that participates in low shear subsolar reconnection will of course be considerably less than that for high shear. Nonetheless, we expect that plasma depletion facilitates reconnection near the subsolar region for low shear and hence subsolar reconnection should not be regarded as restricted to high shear conditions.

To test this observationally, we calculate B_r using $B(1)$ and B_m and calculate B_r^* using $B(2)$ and B_m. That is,

$$B_r = B(1)\frac{a - \cos(\Delta\theta)}{\sqrt{1 + a^2 - 2a\cos(\Delta\theta)}}, \quad a = \frac{B(1)}{B_m} \tag{11a}$$

$$B_r^* = B(2)\frac{a - \cos(\Delta\theta)}{\sqrt{1 + a^2 - 2a\cos(\Delta\theta)}}, \quad a = \frac{B(2)}{B_m} \tag{11b}$$

Figure 6 shows the ratios $B_r/B(1)$ and $B_r^*/B(1)$ versus $\Delta\theta$. The curves are B_r/B_s calculated from (5) assuming that $B_m/B_s = 1$, 2, and 10. The top panel shows the $S(\Delta\theta)$ one might infer by excluding the effects of plasma depletion. The data closely follow the $B_m/B_s = 2$ curve, and the low shear events have $B_r < 0$. The bottom panel shows the $S(\Delta\theta)$ that is actually observed, since by using $B(2)$ and B_m we have included the field increase in the PDL. In contrast to the top panel, most of the low shear events have $B_r^* > 0$, indicating that component merging does indeed remain allowed for low shear. Thus the observations indicate that we should use an $S(\Delta\theta)$ which does not become negative for $\cos(\Delta\theta) > 0.5$. We therefore adopt the simplest form which gives $B_r = 0$ for $\Delta\theta = 0$: $S(\Delta\theta) = \Delta\theta/\pi$. This relation is used below to remove the general dependence of $\beta(2)/\beta(1)$ on magnetic shear.

This result, that B_r is positive even for low shear with B_r monotonically decreasing as the shear gets smaller, has significant implications for solar wind–magnetosphere interactions when the IMF is northward. The IMF influence on the magnetosphere is often regarded as a half-wave rectifier in the sense that the interaction shuts off when the IMF turns northward [e.g., *Crooker et al.*, 1980]. Our result implies that although the strongest interaction occurs for southward IMF, it is not correct to assume that the interaction disappears for northward IMF. Specifically, it means that some subsolar merging may occur for northward IMF, as in fact reported by *Paschmann et al.* [1990], and drive convection in the same sense as the dominant pattern when the IMF is southward. Conventionally, this residual convection for $B_z > 0$ has been attributed to a viscous interaction [e.g., *Reiff*, 1984], but our result suggests that the residual convection may be due to the relatively small degree of subsolar reconnection persisting for $B_z > 0$.

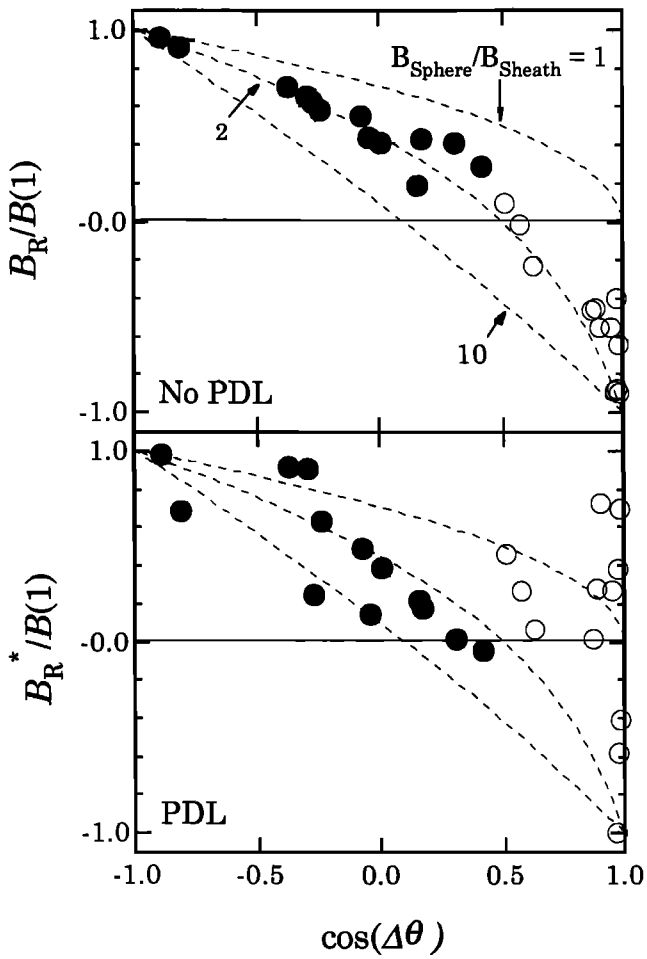

Figure 6. Plot of the magnetosheath field component available for component reconnection, B_r, calculated in two ways and divided by the upstream magnetosheath field, $B(1)$, versus magnetic shear. Upper panel shows B_r calculated using the magnetic field in the magnetosheath proper. Bottom panel shows B_r^* calculated using the magnetosheath field adjacent to the magnetopause.

Reconnection Efficiency

To examine the data for possible dependence on reconnection efficiency, we use (8) to calculate D/k, using $S(\Delta\theta) = \Delta\theta/\pi$, removing the general effects of magnetic shear, field strength, and dynamic pressure and leaving only effects due to changes in k. If k is constant, the same value of D/k for the CCE and IRM cases should give the same value of $\beta(2)/\beta(1)$. Figure 7 shows $\beta(2)/\beta(1)$ plotted versus D/k. Although $\beta(2)/\beta(1)$ generally increases with increasing D/k as we expect, the CCE data are displaced to lower $\beta(2)/\beta(1)$ relative to the IRM data for the same value of D/k. Linear regression of $\log(\beta(2)/\beta(1))$ with $\log(D/k)$ gives $\beta(2)/\beta(1) = g(D/k)^h$, with $g = 1.05$, $h = 0.35$, and a regression coefficient, $r = 0.70$, for IRM and $g = 0.35$, $h = 0.43$, and $r = 0.66$ for CCE. This result implies either that the MHD flow is significantly different for the CCE events or that the reconnection efficiency is lower for the CCE events.

It is unlikely that there is a large difference in the character of the magnetosheath flow around the magnetosphere between the CCE and IRM events. This is because the di-

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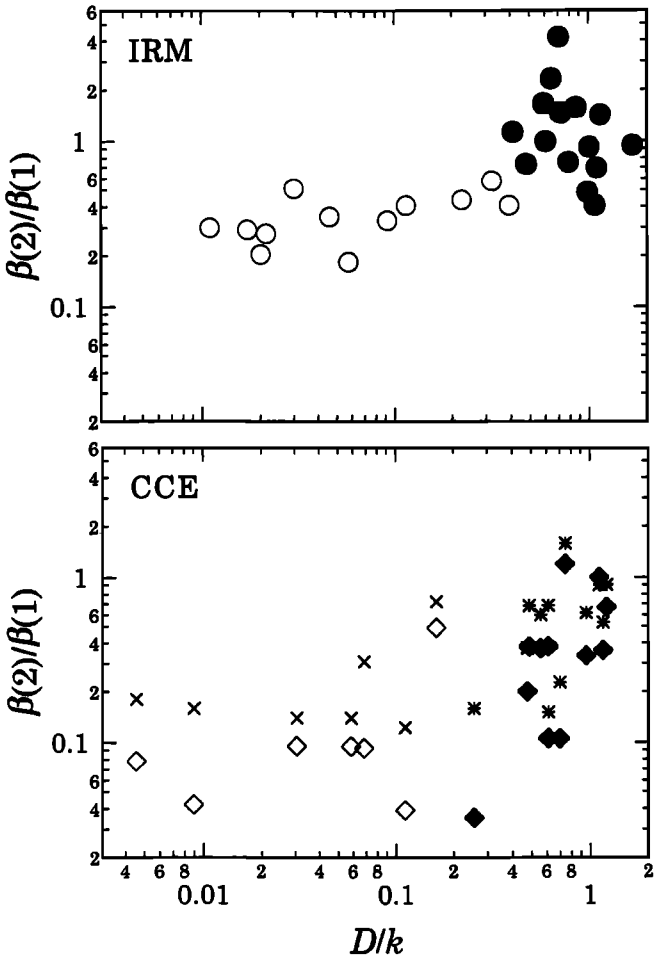

Figure 7. Plot of $\beta(2)/\beta(1)$ plotted versus D/k, where D is the ratio of reconnection to solar wind electric field, that is, $D = E_r/E_{sw}$, and k is the reconnection efficiency. The IRM and CCE events span the same range of D/k while the CCE values for $\beta(2)/\beta(1)$ are systematically a factor of 3 lower than $\beta(2)/\beta(1)$ for the same D/k.

mensions of the system scale as $P_{ram}^{1/6}$ and the magnetopause and bow shock remain self-similar for different dynamic pressures [*Roelof and Sibeck*, 1993; *Peredo et al.*, 1995]. The thickness of the magnetosheath relative to the magnetopause standoff distance does not change with P_{ram} and is only weakly dependent on the solar wind Mach number. For a change in M_{Asw} from 2 to 8, the relative thickness of the subsolar magnetosheath changes by +5%; that is, the subsolar magnetosheath relative thickness increases slightly [*Peredo et al.*, 1995]. Thus the magnetosheath cross-sectional area relative to the obstacle size should be larger for the CCE than for the IRM events, so changes in MHD flow should make the PDL more prominent for the IRM events, opposite to what we observe. Hence geometry effects do not explain the observed $\beta(2)/\beta(1)$ difference between IRM and CCE. The results therefore imply $k_{CCE} < k_{IRM}$; the reconnection efficiency must be lower for the CCE events.

5. Discussion

The comparison of IRM and CCE data sets shows that the apparent discrepancy between them, the absence (presence) of a depletion layer for high shear in the IRM (CCE) data, can be understood as reflecting the difference in balance between the imposed solar wind electric field and the subsolar reconnection electric field. Both data sets display an increase in plasma depletion with decreasing magnetic shear consistent with our model, which predicts the same shear dependence for fixed values of other parameters. The diminution of plasma depletion with increasing sheath field strength for the high shear CCE events is also predicted by our model and results from an increase in the magnetosheath Alfvén velocity and hence in the reconnection rate. The difference in pressure between IRM and CCE events is not sufficient to account for the different in PDL prevalence, however, and indicates that the reconnection efficiency is lower for the CCE than for the IRM events.

To estimate the difference between k_{CCE} and k_{IRM}, we consider how much the CCE data must be moved to the left in Figure 7 so that it makes a good overlap with the IRM data. The regression coefficient between $\log(\beta(2)/\beta(1))$ and $\log(D/k)$ for the combined data set is 0.57. If we allow $k(\beta)$ to have the form $k(\beta) = k_0/(1 + \beta/\beta_c)$ where k_0 and β_c are constants, we can get a rough estimate for β_c by applying the factor $1/(1 + \beta/\beta_c)$ and adjusting β_c to obtain a maximum regression coefficient in the combined data. This form is arbitrary and was chosen simply because it is the simplest form that gives $k \rightarrow 0$ for large β and has no effect on k for small β. Using $\beta_c = 10$, 2.0, and 0.2 gives $r = 0.66$, 0.74, and 0.78, respectively. Values of $\beta_c < 0.2$ do not change the regression coefficient, probably indicating that the reconnection efficiency first begins to decrease for β smaller than the β values sampled by IRM. Figure 8 shows $\beta(2)/\beta(1)$ plotted versus $(D/k)/(1 + \beta(1)/0.2)$, showing that by allowing for some

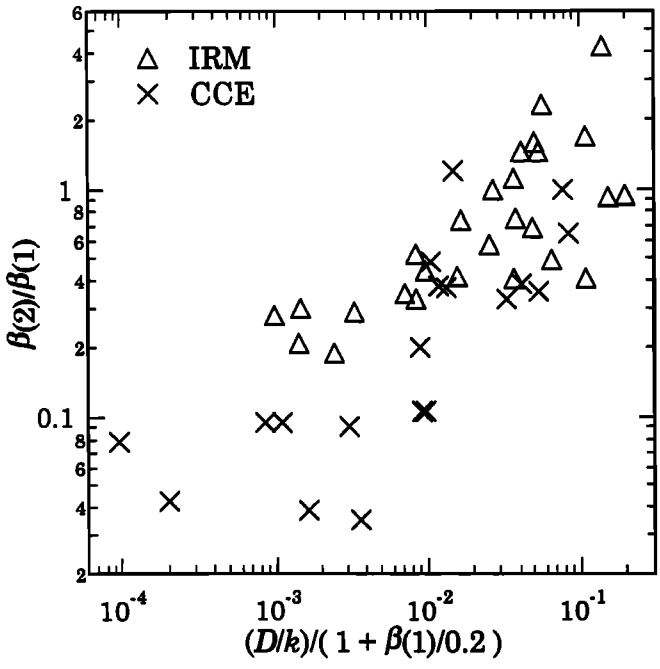

Figure 8. Plot of $\beta(2)/\beta(1)$ versus $D/k/(1 + \beta(1)/0.2)$ showing that the difference in plasma depletion between CCE and IRM can be reconciled if the reconnection efficiency, k, is approximately inversely proportional to β.

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decrease in k with β, one can bring the CCE and IRM data to follow similar trends in $\beta(2)/\beta(1)$. Assuming then that k first begins to decrease for $\beta \sim 1$, and given that $k \sim 1/\beta$ brings the CCE and IRM $\beta(2)/\beta(1)$ variations approximately into agreement, we infer that the ratio of the average reconnection efficiencies for CCE and IRM is roughly given by $k_{CCE}/k_{IRM} \sim \beta(1)_{IRM}/\beta(1)_{CCE} = 2.5/8 = 0.3$.

Previous reports have also found evidence linking increasing β to decreased reconnection efficiency. *Paschmann et al.* [1986] found that the occurrence of reconnection flows is anticorrelated with magnetosheath β and that the magnitude of the reconnection flows decreased as β increased. Using geomagnetic activity as a proxy for reconnection, *Scurry et al.* [1994] inferred that the reconnection efficiency was lower when β was high in the magnetosheath. *Phan et al.* [1996a] also found that reconnection signatures are less prevalent when the magnetosheath β is high, although they point out that direct observation of reconnection flows is more problematical under high β conditions. In the present study we have used the degree of plasma depletion as an indicator of the magnetopause flow boundary condition. This technique therefore provides independent confirmation that the reconnection efficiency decreases with increasing magnetosheath β. Moreover, we have found that the efficiency decreases roughly inversely proportionally to β and that the decrease probably begins for $\beta \sim 1$.

The results imply that reconnection and plasma depletion are closely coupled dynamically. Specifically, when the upstream conditions are unfavorable for reconnection, plasma depletion proceeds and changes conditions near the magnetopause to facilitate some subsolar reconnection. Plasma depletion changes three factors affecting reconnection, and all of them change so as to increase the subsolar reconnection rate. First, the increased magnetic field strength in the PDL implies that the component of magnetosheath field available for reconnection increases. Very significantly, this effect implies that reconnection may remain possible even for very low magnetic shear (see Figure 5). Second, the increased magnetic field and decreased density in the PDL facilitate reconnection by increasing the Alfvén velocity near the magnetopause. Third, the decreased β in the PDL implies that the reconnection efficiency with the plasma in the PDL is higher than it would be with the plasma in the magnetosheath proper. We therefore suggest that the PDL may play an important role in changing the conditions near the subsolar magnetopause to promote subsolar reconnection.

It is clear, however, that the effect of PDL formation is not so great that reconnection flows fully take up the imposed solar wind flow. For high shear and moderate β conditions the subsolar reconnection flows apparently do keep up with the imposed solar wind flow. The simple fact that the PDL forms under low shear and/or high β conditions shows that the reconnection flow does not keep up with the solar wind flow, despite the effect of the PDL to enhance subsolar reconnection. It is clear from Figure 5 that the reconnection rate for high shear will always exceed that for low shear even when the PDL does form. Thus, in our model, PDL formation occurs in response to reductions in the subsolar reconnection rate relative to the solar wind electric field, and the effect of PDL formation on reconnection is to allow somewhat more subsolar reconnection for moderate shear than would otherwise occur and most importantly allow some subsolar reconnection for low shear when it is impossible otherwise.

It is conventionally assumed that subsolar reconnection does not occur for northward IMF [*Sonnerup*, 1984; *Reiff*, 1984], and this is motivated by the half-wave rectifier analogy for subsolar reconnection which results from the antiparallel merging hypothesis [*Crooker*, 1980]. Under this assumption polar cap convection, which displays a so-called four-cell pattern, two poleward cells giving sunward convection at the center of the polar cap and two equatorward cells giving sunward convection at the equatorward edge of the auroral zone, is understood in terms of the combined effects of cusp reconnection and a viscous interaction [*Maezawa*, 1976; *Reiff*, 1984]. Cusp reconnection is thought to drive the poleward cells, while the viscous interaction drives the equatorward cells [*Burke et al.*, 1979; *Burch et al.*, 1980]. However, owing to the effects of plasma depletion, we expect that subsolar merging does indeed occur for northward IMF, and hence the assumption underlying the conventional understanding of convection for northward IMF may be inaccurate. The low shear magnetopause displays convection and electric field structures consistent with a rotational discontinuity [*Paschmann et al.*, 1990], the subsolar magnetopause appears to generally exhibit a reconnection topology even for northward IMF [*Fuselier et al.*, 1995], and observations of ion velocity dispersion in the subsolar region do indeed indicate that reconnection occurs equatorward of the cusp for northward IMF [*Onsager and Fuselier*, 1994; *Fuselier et al.*, 1997]. Since the equatorward convection cells are in the same sense as convection driven by subsolar reconnection, these cells could be driven by the residual subsolar reconnection facilitated by the PDL. We therefore propose that the four-cell convection pattern observed for northward IMF can be understood as resulting from the combined effects of cusp reconnection and simultaneous weak subsolar reconnection without recourse to a viscous interaction.

That a PDL forms even for high shear indicates that the reconnection electric field fails to keep up with the imposed solar wind flow under high β conditions. We interpret this saturation in terms of a reconnection efficiency which falls off approximately like $1/\beta$ for β larger than about 1. This may account for the observed saturation of the polar cap potential that occurs during periods of high geomagnetic activity [*Wygant et al.*, 1983]. *Reiff et al.* [1981] also noted that the polar cap potential reached a maximum value during active times, and inferred that to achieve the observed convection, the subsolar magnetic field had to be enhanced over its nominal magnetosheath value; that is, a PDL must have formed. As discussed above, PDL formation is intimately related to a reduction of the reconnection rate relative to the solar wind electric field. We find that the reconnection efficiency is reduced under high β, high shear conditions and that this results in PDL formation. The conclusions of Reiff et al. and Wygant et al. therefore reflect the same underlying phenomenon and are mutually consistent.

It is significant that high β rather than high solar wind ram pressure may be principally responsible for causing PDL formation for the high shear CCE events. The magnetosheath β is $2\mu_0 n_{sh} k T_{sh}/B_{sh}^2$, and for quasi-perpendicular shocks in the strong shock limit, T_{sh} is given by

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$$T_{sh} = \frac{2(\gamma - 1)}{(\gamma + 1)^2} m V_{sw}^2 \quad (12)$$

where γ is the ratio of specific heats. Using this together with the compression relations for B_{sh} and n_{sh} we have

$$\beta_{sh} = \frac{4(\gamma - 1)}{\eta(\gamma + 1)^2} \frac{\mu_0 m n_{sw}}{B_{sw}^2} V_{sw}^2 \quad (13)$$

Using $\gamma = 5/3$ and $\eta = (\gamma + 1)/(\gamma - 1) = 4$ and recognizing that the second quotient is the square of the solar wind Alfvén speed, we have

$$\beta_{sh} = \frac{3}{32} M_{Asw}^2 \quad (14)$$

Since $M_{Asw}^2 \propto n_{sw} V_{sw}^2 / B_{sw}^2$, it is clear that it is not the solar wind ram pressure alone which principally determines the downstream β. We expect that M_{Asw} rather than ram pressure should be correlated best with low reconnection efficiency and PDL formation. One could then very well have high solar wind ram pressure conditions without a significant plasma depletion layer if the solar wind magnetic field is also enhanced.

6. Conclusions

In this paper we examined the physical mechanisms leading to the different prevalence of plasma depletion in CCE and IRM observations. The ratio of the reconnection electric field to the solar wind electric field, $D = E_r/E_{sw}$, measures the relative rate of magnetic flux transfer by reconnection to the solar wind inflow. Since plasma depletion corresponds to a reconnection flow lower than that required to take up the imposed solar wind flow, this ratio serves as a useful indicator of plasma depletion: $D = 0$ indicates plasma depletion whereas $D \sim 1$ corresponds to no plasma depletion. By expressing D in terms of solar wind or magnetosheath parameters, one can identify the physical variables that influence the occurrence of plasma depletion (see (7) and (8)). Plasma depletion is most sensitive to the magnetic shear at the magnetopause and is most likely for low shear. The model also predicts that plasma depletion decreases as the magnetosheath or solar wind magnetic field strength increases. Increased solar wind dynamic pressure should correspond to increased depletion as well. The reconnection efficiency also has a strong effect on plasma depletion such that plasma depletion should be more prevalent for high magnetosheath β.

The predictions of this model were tested by combining the databases of magnetosheath observations from the CCE and IRM spacecraft. All of the model predictions were borne out observationally, indicating that this simple description captures the essential factors influencing PDL formation. We find that the reconnection efficiency must have been about a factor of three lower for the CCE events than for the IRM events, most probably owing to the threefold higher β for the CCE events. We therefore infer that the reconnection efficiency is roughly inversely proportional to β for β greater than about 1. This conclusion is based on changes in the magnetosheath flow boundary condition as evidenced in the prevalence of plasma depletion. The result provides independent confirmation of the inverse relationship between reconnection efficiency and magnetosheath β predicted by *Sonnerup* [1974], found in direct observations of accelerated flow at the magnetopause [*Paschmann et al.*, 1986; *Phan et al.*, 1996a], and inferred by variations in geomagnetic activity with upstream conditions [*Scurry et al.*, 1994].

The decrease in reconnection efficiency with β implies that magnetosheath β larger than about 5 or so should correlate with the occurrence of plasma depletion for southward IMF. We therefore expect plasma depletion for high magnetic shear to occur primarily during high solar wind density conditions because although high solar wind velocities produce large increases in solar wind pressure, they do not produce magnetosheath β as high as those observed during high solar wind density conditions.

Because plasma depletion becomes more pronounced for low shear, the magnetosheath magnetic field imposed on the magnetopause is not independent of the shear angle, but increases as the sheath field turns northward. This implies that component merging should remain allowed in the subsolar region for all IMF orientations except purely northward. Plasma depletion also promotes reconnection by increasing the subsolar magnetosheath Alfvén speed and decreasing the subsolar magnetosheath β, thus increasing the reconnection efficiency. The effects of plasma depletion therefore facilitate some subsolar reconnection for northward IMF.

It is conventionally assumed that subsolar reconnection does not occur for northward IMF, and under this assumption four-cell polar cap convection, which occurs for IMF $B_z > 0$, is understood in terms of the combined effects of cusp reconnection and a viscous interaction. Our results suggest that the effects of plasma depletion promote some residual subsolar reconnection even for northward IMF. We therefore suggest that four-cell polar cap convection can be understood as resulting from the combined effects of cusp reconnection and simultaneous weak subsolar reconnection without recourse to a viscous interaction.

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References

Acuña, M. H., G. W. Ousley, R. W. McEntire, D. Bryant, and G. Paschmann, Editorial: AMPTE-Mission Overview, *IEEE Trans. Geosci. Remote Sens.*, *23*, 175, 1985.

Anderson, B. J., and S. A. Fuselier, Magnetic pulsations from 0.1 to 4.0 Hz and associated plasma properties in the Earth's subsolar magnetosheath and plasma depletion layer, *J. Geophys. Res.*, *98*, 1461, 1993.

Anderson, B. J., S. A. Fuselier, S. P. Gary, and R. E. Denton, Magnetic spectral signatures from 0.1 to 4.0 Hz in the Earth's magnetosheath and plasma depletion layer, *J. Geophys. Res.*, *99*, 5877, 1994.

Burch, J. L., P. H. Reiff, R. A. Heelis, R. W. Spiro, and S. A. Fields, Cusp region particle precipitation and ion con-

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vection for northward interplanetary magnetic field, *Geophys. Res. Lett.*, *7*, 393–396, 1980.

Burke, W. J., M. C. Kelley, R. C. Sagalyn, M. Smiddy, and S. T. Lai, Polar cap electric field structures with a northward interplanetary magnetic field, *Geophys. Res. Lett.*, *6*, 21–24, 1979.

Crooker, N. U., The half-wave rectifier response of the magnetosphere and anti-parallel merging, *J. Geophys. Res.*, *85*, 575–578, 1980.

Denton, R. E., S. P. Gary, B. J. Anderson, S. A. Fuselier, and M. K. Hudson, Low-frequency magnetic fluctuation spectra in the magnetosheath and plasma depletion layer, *J. Geophys. Res.*, *99*, 5893, 1994.

Fuselier, S. A., B. J. Anderson, and T. G. Onsager, Particle signatures of magnetic topology at the magnetopause: AMPTE/CCE observations, *J. Geophys. Res.*, *100*, 11,805, 1995.

Fuselier, S. A., B. J. Anderson, and T. G. Onsager, Electron and ion signatures of field line topology at the low shear magnetopause, *J. Geophys. Res.*, in press, 1997.

Gary, S. P., M. E. McKean, D. Winske, B. J. Anderson, R. E. Denton, S. A. Fuselier, Proton cyclotron anisotropy instability and the anisotropy/beta inverse correlation, *J. Geophys. Res.*, *99*, 5903, 1994.

Kennel, C. F., J. P. Edmiston, and T. Hada, A quarter century of collisionless shock research, in *Collisionless Shocks in the Heliosphere: A Tutorial Review, Geophys. Monogr. Ser.*, vol. 34, edited by R. G. Stone and B. T. Tsurutani, pp. 1–36, AGU, Washingtin, D. C., 1985.

Lees, L., Interaction between the solar plasma wind and the geomagnetic cavity, *AIAA J.*, *2*, 2065, 1964.

Lin, N., M. J. Engebretson, R. L. McPherron, M. G. Kivelson, W. Baumjohann, H. Luehr, T. A. Potemra, B. J. Anderson, and L. J. Zanetti, A comparison of ULF fluctuations in the solar wind, magnetosheath, and dayside magnetosphere, 2, Field and plasma conditions in the magnetosheath, *J. Geophys. Res.*, *96*, 3455, 1991.

Luhmann, J. G., C. T. Russell, and R. C. Elphic, Spatial distributions of magnetic field fluctuations in the dayside magnetosheath, *J. Geophys. Res.*, *91*, 1711, 1986.

Maezawa, K., Magnetic convection induced by the positive and negative *z*-components of the interplanetary magnetic field: Quantitative analysis using polar cap magnetic records, *J. Geophys. Res.*, *81*, 2289, 1976.

Midgley, J. E., and L. Davis, Calculation by a moment technique of the perturbation of the geomagnetic field by the solar wind, *J. Geophys. Res.*, *68*, 5111, 1963.

Onsager, T. G., and S. A. Fuselier, The location of magnetic reconnection for northward and southward interplanetary magnetic field, in *Solar System Plasma Physics: Resolution of Processes in Space and Time, Geophys. Monogr. Ser.*, edited by J. L. Burch and J. H. Waite Jr., p. 183, AGU, Washington D. C., 1994.

Paschmann, G., I. Papamastorakis, W. Baumjohann, N. Sckopke, C. W. Carlson, B. U. O. Sonnerup, and H. Lühr, The magnetopause for large magnetic shear: AMPTE/IRM observations, *J. Geophys. Res.*, *91*, 11,099–11,115, 1986.

Paschmann, G., B. U. O. Sonnerup, I. Papamastorakis, W. Baumjohann, N. Sckopke, and H. Lühr, The magnetopause and boundary layer for small magnetic shear: Convection electric fields and reconnection, *Geophys. Res. Lett.*, *17*, 1829–1832, 1990.

Peredo, M., J. A.. Slavin, E. Mazur, and S. A. Curtis, Three-dimensional position and shape ot the bow shock and their variation with Alfvénic, sonic and magnetosonic Mach numbers and interplanetary magnetic field orientation, *J. Geophys. Res.*, *100*, 7907–7916, 1995.

Phan, T.-D., G. Paschmann, W. Baumjohann, N. Sckopke, and H., Lühr, The magnetosheath region adjacent to the dayside magnetopause: AMPTE/IRM observations, *J. Geophys. Res.*, *99*, 121, 1994.

Phan, T.-D., G. Paschmann, and B. U. O. Sonnerup, Low-latitude dayside magnetopause and boundary layer for high magnetic shear, 2, Occurrence of magnetic reconnection, *J. Geophys. Res.*, *101*, 7817–7828, 1996a.

Phan, T.-D., et al., The subsolar magnetosheath and magnetopause for high solar wind ram pressure: WIND observations, *Geophys. Res. Lett.*, *23*, 1279–1282, 1996b.

Reiff, P. H., R. W. Spiro, and T. W. Hill, Dependence of polar cap potential drop on interplanetary parameters, *J. Geophys. Res.*, *86*, 7639, 1981.

Reiff, P. H., Evidence of magnetic merging from low-altitude spacecraft and ground-based experiments, in *Magnetic Reconnection in Space and Laboratory Plasmas, Geophys. Monogr. Ser.*, vol. 30, edited by E. W. Hones, pp. 105–113, AGU, Washington, D. C., 1984.

Roelof, E. C., and D. G. Sibeck, Magnetopause shape as a bivariate function of interplanetary magnetic field B_z and solar wind dynamic pressure, *J. Geophys. Res.*, *98*, 21,421, 1993.

Scurry, L., C. T. Russell, and J. T. Gosling, Geomagnetic activity and the beta dependence of the dayside reconnection rate, *J. Geophys. Res.*, *99*, 14,811, 1994.

Sonnerup, B. U. O., Magnetopause reconnection rate, *J. Geophys. Res.*, *79*, 1546, 1974.

Sonnerup, B. U. O., Magnetic field reconnection at the magnetopause: an overview, in *Magnetic Reconnection in Space and Laboratory Plasmas, Geophys. Monogr. Ser.*, vol. 30, edited by E. W. Hones, p. 92, AGU, Washington, D. C., 1984.

Wu, C. C., MHD flow past an obstacle: Large-scale flow in the magnetosheath, Geophys. Res. Lett., *19*, 87–90, 1992.

Wygant, J. R., R. B. Torbert, and F. S. Mozer, Comparison of S3-3 polar cap potential drops with the interplanetary magnetic field and models of magnetopause reconnection, *J. Geophys. Res.*, *88*, 5727, 1983.

Zwan, B. J., and R. A. Wolf, Depletion of solar wind plasma near a planetary boundary, *J. Geophys. Res.*, *81*, 1636, 1976.

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